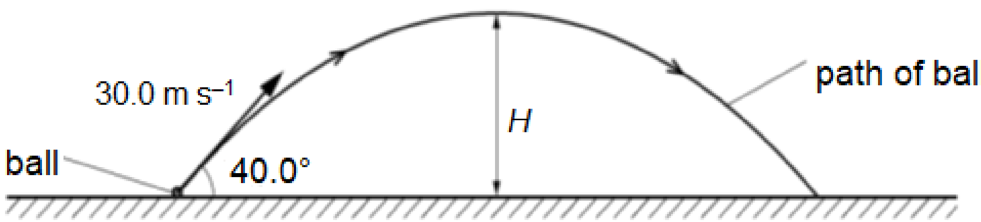
Answer **all** the questions in this Section in the spaces provided.

1	**(a)**		Define *acceleration*.	[1]
	(a)		**Solution:** It is the change in velocity per unit time taken. or It is the rate of change of velocity.	**1**
	(b)		A ball of mass 400 g is thrown with an initial velocity of 30.0 m s^{-1} at an angle of 40.0° to the horizontal, as shown in Fig. 1.1.  **Fig. 1.1** (not to scale) Air resistance is negligible. The ball reaches a maximum height *H* after a time of 1.97 s.	
		(i)	Calculate	
			1. the initial kinetic energy of the ball, kinetic energy = ... J	[2]
		1.	**Solution:** $\text{K.E.} = \frac{1}{2}mv^2$ $= \frac{1}{2} \times 0.4 \times 30^2$ $= 180$ J	**1** **1**
			2. the maximum height *H* of the ball,	

			H = ………………………………… m	[2]
		2.	**Solution:** Taking downwards as positive, **Time taken to reach maximum height is the same as time taken for ball to go from maximum height back to ground level.** $s_y = u_y t + \frac{1}{2}gt^2$ $s_y = 0 + \frac{1}{2} \times 9.81 \times (1.97)^2$ $= 19.0$ m Or For object moving to max height from origin. $s_y = u_y t + \frac{1}{2}gt^2$ $s_y = 30\sin40(1.97) + \frac{1}{2} \times (-9.81) \times (1.97)^2$ $= 19.0$ m Or $v_y^2 = u_y^2 + 2a_y s_y$ $0 = (30\sin40)^2 + 2(-9.81)s_y$ $s_y = 18.95 = 19.0$ m	**1** **1** **1** **1** **1** **1**
			3. the gravitational potential energy of the ball at height H. potential energy = ………………………………… J	[1]
		3.	**Solution:** G.P.E. = mgh $= 0.4 \times 9.81 \times 19.0$ $= 74.6$ J **Allow for ecf of h from bi2.**	**1**
		(ii)	**1.** Determine the kinetic energy of the ball at its maximum height.	

			kinetic energy = ………………………………. J	[1]
		1.	**Solution:** K.E. = Initial K.E. - G.P.E. = 180 – 74.6 = 105 J	1
			2. Explain why the kinetic energy of the ball at the maximum height is not zero. …………………………………………………………………………… ……………………………………………………………………………	[1]
		2.	**Solution:** The horizontal velocity is not zero and the object is still moving horizontally. Thus it has K.E.	1
		(iii)	Determine the maximum horizontal distance travelled by the ball. horizontal distance = ………………………………. m	[2]
		(iii)	**Solution:** $s_x = u_x t$ $= (30.0 \cos 40^0)(1.97 \times 2)$ $= 90.5$ m Note: Time taken for object to go to max height is the same as time taken to go from max height to ground.	1 1

2	A salvage crane with an output power of 522 kW is lifting a fully submerged cube crate	